

IT1223(P) - Database Management Systems (P)
Department of Physical Science

University of Vavuniya

Lab Sheet - 03

Answers

Q7) Using students database you have created in the previous exercise do the following:

IN

- I. Using IN operator in MYSQL to list all students with a course of "Computer Science" or "Physical Science" from the table fasstudents.**

query

- **SELECT * FROM fasstudents WHERE course IN ('Computer Science','Physical Science');**

```
Database changed
MariaDB [it1223p]> SELECT * FROM fasstudents WHERE course IN ('Computer Science','Physical Science');
+-----+-----+-----+-----+-----+-----+-----+
| FirstName | LastName | Address | Sex | Date_of_Birth | Course | Telephone |
+-----+-----+-----+-----+-----+-----+-----+
| Nimal | Perera | 10A, Galle Road,Colombo | M | 1990-05-15 | Computer Science | 011-2345678 |
| Anula | Rajapakse | 5B, Matara Street, Matara | F | 1995-12-03 | Computer Science | 041-5678901 |
| Sujatha | Bandara | 20, Jaffna Road, Jaffna | F | 1991-11-25 | Computer Science | 021-7890123 |
| Lakmini | Wijesinghe | 8C, Badulla Street,Badulla | F | 1989-06-30 | Computer Science | 055-9012345 |
| Upul | Dissanayake | 40, Ampara Road,Ampara | M | 1996-02-14 | Computer Science | 063-0123456 |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)
```

- II. Using BETWEEN condition in MYSQL list all students born between '1990-01-31' AND '1991-01-31' from the table fasstudents**

query

- **SELECT * FROM fasstudents WHERE Date_of_Birth BETWEEN '1990-01-31' AND '1991-01-31';**

```
MariaDB [it1223p]> SELECT * FROM fasstudents WHERE Date_of_Birth BETWEEN '1990-01-31' AND '1991-01-31';
+-----+-----+-----+-----+-----+-----+-----+
| FirstName | LastName | Address | Sex | Date_of_Birth | Course | Telephone |
+-----+-----+-----+-----+-----+-----+-----+
| Nimal | Perera | 10A, Galle Road,Colombo | M | 1990-05-15 | Computer Science | 011-2345678 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```

GROUP BY

III. Using GROUP BY in MYSQL to count the number of students following Computer Science and Physical Science course.

query

- CREATE TABLE employee_tbl1(
employee_id int,
name VARCHAR(50),
work_date DATE,
daily_typing_pages int(5));

```
MariaDB [it1223p]> CREATE TABLE employee_tbl1(  
-> employee_id int,  
-> name VARCHAR(50),  
-> work_date DATE,  
-> daily_typing_pages int(5));  
Query OK, 0 rows affected (0.214 sec)
```

- INSERT INTO employee_tbl1(employee_id,name,work_date,daily_typing_pages)
VALUES (1,'JOHN','2007-01-24',250),

```
(2,'RAM','2007-05-27',220),
(3,'JACK','2007-05-06',170),
(3,'JACK','2007-05-06',100),
(4,'JILLY','2007-04-06',220),
(5,'ZARA','2007-06-06',300),
(5,'JACK','2007-02-06',350);
```

```
MariaDB [it1223p]> INSERT INTO employee_tbl1(employee_id,name,work_date,daily_typing_pages)
-> VALUES (1,'JOHN','2007-01-24',250),
-> (2,'RAM','2007-05-27',220),
-> (3,'JACK','2007-05-06',170),
-> (3,'JACK','2007-05-06',100),
-> (4,'JILLY','2007-04-06',220),
-> (5,'ZARA','2007-06-06',300),
-> (5,'JACK','2007-02-06',350);
Query OK, 7 rows affected (0.042 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

- SELECT * FROM employee_tbl;

```
MariaDB [it1223p]> SELECT * FROM employee_tbl1;
+-----+-----+-----+-----+
| employee_id | name  | work_date | daily_typing_pages |
+-----+-----+-----+-----+
|          1 | JOHN  | 2007-01-24 |          250 |
|          2 | RAM   | 2007-05-27 |          220 |
|          3 | JACK  | 2007-05-06 |          170 |
|          3 | JACK  | 2007-05-06 |          100 |
|          4 | JILLY | 2007-04-06 |          220 |
|          5 | ZARA  | 2007-06-06 |          300 |
|          5 | JACK  | 2007-02-06 |          350 |
+-----+-----+-----+-----+
7 rows in set (0.000 sec)
```

- SELECT Course,COUNT(*)AS num_students FROM fasstudents WHERE Course IN ('Computer Science','Physical Science') GROUP BY Course;

```
MariaDB [it1223p]> SELECT Course,COUNT(*)AS num_students FROM fasstudents WHERE Course IN ('Computer Science','Physical Science') GROUP BY Course;
```

Course	num_students
Computer Science	5

```
1 row in set (0.001 sec)
```

- IV. Based on the above table if we want to count number of days each employee did work how can you write an SQL Query?**

query

- `SELECT employee_id,SUM(work_date)AS total_work_days FROM employee_tbl1 GROUP BY employee_id;`

```
MariaDB [it1223p]> SELECT employee_id,SUM(work_date)AS total_work_days FROM employee_tbl1 GROUP BY employee_id;
```

employee_id	total_work_days
1	20070124
2	20070527
3	40141012
4	20070406
5	40140812

```
5 rows in set (0.001 sec)
```

Cloning a table

- V. Explain how you can create an exact copy(clone) of the table fasstudents. Name the newly created table as clonefasstudents.**

query

`CREATE TABLE clonefasstudents AS SELECT * FROM fasstudents;`

```
MariaDB [it1223p]> CREATE TABLE clonefasstudents AS SELECT * FROM fasstudents;
Query OK, 10 rows affected (0.333 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

Q8) Backup and Restore MYSQL Database

- VI. vi. Create a table in your Ubuntu desktop and backup your students database as studentsbackup.sql.**

query

```
mysqldump -u username -p it1223p > studentsbackup.sql;
```

- VII. How can you restore the database with a new name?**

query

- CREATE DATABASE restored_stdents;
- mysqldump -u username -p restored_stdents < studentsbackup.sql